D52 | Three answers, different questions

Fictional SaaS teaching case | INR crore | 13 November 2026 | Build your own model first; this authored workbook is a checking aid, not a market quote.

Separate the questions	VC is a conditional investor-specific entry ceiling. Comps are invented current market cross-checks. DCF is a separate operating forecast. Do not average or sum these outputs.
Scope and exclusions	Current ARR 8, debt 3, surplus cash 1. No other assets, senior claims, preferences or current dilution. A raise of 10 enters only the VC ownership calculation, not operating FCFF.
VC bridge	Exit EV 800 less debt 30 plus surplus cash 10 = equity 780. Retention 60% and terminal gross hurdle 10x imply post-money ceiling 46.8 and pre-money 36.8. Six years is not an extra exponent on 10x.
Matched comparable basis	Current EV/current ARR multiples 6x and 8x use current ARR 8. Deduct current net debt 2 once for pre-money equity 46 and 62. Fictional comps are not actual Indian transaction evidence.
Operating DCF	Year-end FCFF -4,-2,1,5,10,16; rate 25%, perpetual growth 3% after year 6. Discount six cash flows and terminal EV at year 6, then deduct current net debt once. No VC exit added.
Controls and limits	Yellow cells are editable. Each method has separate validity checks. A mathematical ceiling can still be below the raise and infeasible. Negative equity is shown, not hidden at zero. Rate <= growth makes perpetuity unavailable.
Source and execution limits	IPEV supports market-participant assumptions, matched multiples, EV/equity allocation and sensitive DCF estimates. Payne scorecard is not forced onto this revenue-bearing case. Native authored-file tests are not an independent learner audit or Excel/Sheets parity.

No real company, trade, external contact, investment recommendation, reporting fair value, legal terms, account action or learner-progress write. Read Practice for controlled tests and source limits.

Inputs | Keep the dates and bases apart

All numbers below are invented assumptions in the original D52 exercise. Source version: 13 November 2026. No real transaction, ARR audit or market-participant calibration has been supplied.

Current ARR	8.00	Current annual recurring revenue, not exit revenue
Current debt	3.00	Measurement-date bridge
Current surplus cash	1.00	Measurement-date bridge
Raise today	10.00	VC ownership only; no follow-on
Exit year	6.00	Fixed six-year timeline; must remain 6
Exit revenue	100.00	Conditional successful year-6 revenue
Exit EV / revenue	8.00	Exit metric basis, not current ARR
Exit debt	30.00	Exit bridge, not current debt
Exit surplus cash	10.00	Exit bridge
Ownership retention	60.00%	60% of purchased ownership remains
Terminal gross hurdle	10.00	10x terminal multiple, NOT 10% annual
Current EV / ARR low	6.00	Invented matched-basis comp
Current EV / ARR high	8.00	Invented matched-basis comp
Annual FCFF rate	25.00%	Risk-adjusted operating discount rate
Perpetual growth	3.00%	Begins after year 6

Year-end FCFF year	**1**	**2**	**3**	**4**	**5**	**6**
FCFF / crore	-4.00	-2.00	1.00	5.00	10.00	16.00

Negative interim FCFF is intentional. Numeric-count and domain checks catch pasted text/blanks in calculation sheets. VC timeline is fixed to 6 years for this exercise; changing the year count requires a new consistent FCFF timeline, not this input alone.

VC | An entry ceiling, not fair value

Conditional successful exit only. Investor makes no follow-on; pro-rata equity receives proceeds. Return hurdle 10x is already a terminal multiple. Retention is applied once.

Valid VC inputs? (1=yes)	1.0000
Exit enterprise value	800.0000
Exit equity value	780.0000
Maximum equity POST-money today	46.8000
Maximum equity PRE-money today	36.8000
Purchased ownership at ceiling	21.37%
Retained exit ownership	12.82%
Investor exit proceeds	100.0000
Gross proceeds / investment	10.0000
Equivalent annual compound hurdle	46.78%
Can raise fit within ceiling?	Feasible arithmetic, not approval

Zero retention gives a zero post-money ceiling and no feasible positive raise; ownership/proceeds are then unavailable. A purchased ownership above 100% is not executable even if the gross-hurdle identity balances. Annual equivalent is diagnostic only, not an extra discount applied to the ceiling.

Comps | Match current EV to current ARR

Two invented comparable indications, not observations of actual Indian sales. Same ARR/accounting/period definitions are stipulated, not independently verified. Current net debt is deducted once.

Measure / crore	Low comp	High comp
Valid current inputs? (1=yes)	1.0000	
Current enterprise value	48.0000	64.0000
Current net debt	2.0000	2.0000
Current equity PRE-money	46.0000	62.0000

No raise proceeds are added to these pre-money indications. Exit revenue 100 is not substituted for current ARR 8. Do not treat a funding-round equity price as an enterprise multiple. Negative equity indications remain visible as a limitation, not an automatic investable value.

A real extension needs transaction dates, price/consideration, debt/cash bridge, metric definitions, accounting basis, growth/risk comparability and retrievable sources. An undisclosed price or ARR remains UNKNOWN, not a fabricated multiple.

DCF | Operating cash first, financing separate

Year-end FCFF belongs to the business, not the investor. Six explicit periods plus a terminal value after year 6. Discount rate and terminal growth must satisfy rate > growth and both > -100%.

Valid DCF inputs? (1=yes)	1.0000

Year-end	**1**	**2**	**3**	**4**	**5**	**6**
FCFF	-4.0000	-2.0000	1.0000	5.0000	10.0000	16.0000
Discount factor	0.8000	0.6400	0.5120	0.4096	0.3277	0.2621
PV of explicit FCFF	-3.2000	-1.2800	0.5120	2.0480	3.2768	4.1943

Terminal EV at year 6	74.9091
PV of terminal EV	19.6370
Sum explicit PV	5.5511
Enterprise value today	25.1881
Current net debt	2.0000
Equity PRE-money today	23.1881

Terminal PV / EV	77.96%

Terminal weight is a diagnostic, not a required percentage or quality score; it can exceed 100% if explicit PV is negative. Negative FCFF or equity values are not floored away. Rate <= growth is unavailable, not zero. Forecast/rate/growth evidence and market-participant calibration are absent.

Sensitivity | Change one basis at a time

Grid values are equity PRE-money in crore. Axes are additional fictional teaching assumptions, not plausible-range estimates or probability-weighted values. The baseline is included in each grid.

VC: exit multiple vs ownership retention; hurdle held at current input			
Exit multiple / retention	**0.4**	**0.6**	**0.8**
6	13.2000	24.8000	36.4000
8	21.2000	36.8000	52.4000
10	29.2000	48.8000	68.4000

DCF: annual rate vs perpetual growth; explicit FCFF held fixed			
Rate / growth	1.00%	3.00%	5.00%
20.00%	34.1288	38.1103	43.1535
25.00%	21.2021	23.1881	25.5712
30.00%	13.4983	14.5990	15.8758

VC terminal hurdle	**8**	**10**	**12**
Pre-money ceiling	48.5000	36.8000	29.0000

Change one input and reconcile the path, not only the final value. Current debt +1 reduces comps/DCF equity by 1 but leaves the VC exit bridge unchanged. Exit debt +1 lowers VC post/pre by retention/hurdle but leaves current comps/DCF unchanged.

Practice | Evidence before a decision

Rebuild independently, then compare formulas and outputs. All numeric inputs, forecast ranges and roles are synthetic; 13 November 2026 is a model-version date, not a company measurement date.

Input registry	Inputs B5:B7 current metric/bridge; B9:B16 VC assumptions; B18:B19 invented comp multiples; B21:B22 rates/growth and G26 terminal FCFF; B26:G26 FCFF. Owner: fictional case author, units crore or ratios, period stated on each sheet.
Controlled change 1	Current debt +1: Comps B10/C10 and DCF B18 fall by 1; VC B10/B11 unchanged. Restore before the next change.
Controlled change 2	Exit debt +1: VC post/pre fall by .06 at baseline; current comps and DCF unchanged. Retention .5 instead of .6 lowers VC post by 7.8. Do not subtract dilution twice.
Failure tests	Try text, blank, zero hurdle, retention outside 0..1, rate=growth, growth=-100%, and a missing FCFF. Invalid method outputs must be unavailable. Zero retention is mathematically valid but infeasible for a positive raise.
Review questions	Why do the three answers differ? Which assumptions explain the disagreement? Which real sources would test current ARR, transaction basis, return/risk/dilution, FCFF and terminal value? Do not average to conceal missing evidence.
Source scope	IPEV 2025 scope, 1.1-1.6, 2.1-2.3, 3.2-3.4 and 3.7 re-read: market participant/date, suitable methods, comparable bases and operating DCF. Equidam is one vendor implementation; its stage success/retention estimates are not adopted here.
Scorecard applicability	Payne/ACA article requires matched local pre/minimal-revenue peers. This established-revenue SaaS case does not supply that benchmark. Critique applicability; do not force a fourth numeric answer or use historical U.S. dollar ranges as current Indian evidence.
Execution record	PASS: 64 baseline checks; 25 controlled-change checks; 10 invalid-input cases; zero-retention/infeasible-raise/negative-equity/reference edges; 64 inserted-row and 64 restored checks. Native LibreOffice against independent rational evaluator, not learner evidence.

Sources: IPEV 2025 valuation guidelines; Equidam, The Logic of the VC Method (30 July 2025); ACA, Bill Payne Scorecard Valuation Methodology (21 October 2019). Exact URLs are in course Study kit. No new external sources or real transactions are represented by the numeric model.